

Volatility Forecasting Using Machine Learning: A Comparative Study of Econometric and Machine Learning Models

Abstract

This study investigates the effectiveness of machine learning methods in forecasting financial market volatility compared to traditional econometric approaches. Using historical market data from Apple Inc. (AAPL), the SPDR S&P 500 ETF (SPY), Microsoft Corporation (MSFT), and the CBOE Volatility Index (VIX), a feature-rich dataset was constructed incorporating lagged returns, realized volatility measures, market sentiment indicators, and trading activity variables. Volatility was estimated using the Garman-Klass estimator and next-day volatility was selected as the prediction target. Baseline forecasting models are first established before evaluating econometric models such as GARCH and EGARCH and machine learning models including Random Forest and XGBoost. The objective is to determine whether machine learning approaches provide superior predictive performance relative to traditional volatility forecasting techniques.

1. Introduction

Volatility forecasting is a fundamental problem in quantitative finance, risk management, portfolio allocation, derivative pricing, and algorithmic trading. Accurate forecasts of future market volatility enable financial institutions and investors to make informed decisions regarding risk exposure and capital allocation.

Traditional volatility forecasting methods such as the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model have been widely adopted due to their ability to capture volatility clustering and persistence. However, advances in machine learning have introduced alternative forecasting approaches capable of modeling nonlinear relationships and complex interactions among financial variables.

This study aims to compare the forecasting performance of traditional econometric models and modern machine learning algorithms using a common dataset and evaluation framework.

Research Question

Can machine learning models outperform traditional econometric models in forecasting next-day equity volatility?

Objectives

- Construct a comprehensive volatility forecasting dataset.
- Establish benchmark forecasting models.
- Evaluate GARCH and EGARCH models.

- Evaluate Random Forest and XGBoost models.
- Compare forecasting performance across model classes.

2. Data Description

Assets

- AAPL
- SPY
- MSFT
- VIX

Period

2016–2025

Data Source

Yahoo Finance (via yfinance)

Target Variable

Next-day Garman-Klass volatility of AAPL.

Features

- Lagged returns
- Rolling volatility measures
- VIX indicators
- SPY market indicators
- MSFT cross-asset indicators
- Trading volume variables

3. Methodology

Volatility Estimation

The Garman-Klass estimator is employed to construct realized volatility from Open, High, Low, and Close prices.

Feature Engineering

Features include return-based predictors, volatility persistence measures, market-wide indicators, and volume-related variables.

Forecasting Framework

The dataset is divided chronologically into training and testing periods to avoid look-ahead bias. Model performance is evaluated using out-of-sample forecasting accuracy.

4. Baseline Models

(To be completed in Day 6)

5. Econometric Models

(To be completed in Day 7)

6. Machine Learning Models

(To be completed in Days 8–9)

7. Results

(To be completed)

8. Conclusion

(To be completed)